

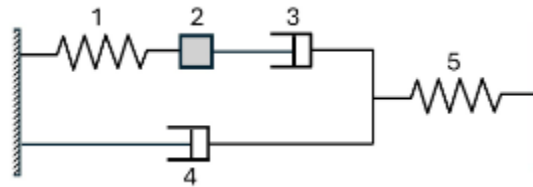
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Professor Luo

Homework 3 Kasmiersky

1. Consider the following diagram with masses, dampers, and springs. The constitutive law constants are K_i , m_i , and b_i , where i stands for the index of the component.



- a. Write out newtons laws without plugging in constitutive laws for this system

$$-f_1 + f_3 = m_2 a_2$$

$$-f_3 - f_4 + f_5 = 0$$

- b. Write out all the kinematic constraint equations.

$$v_1 + v_3 = v_4$$

$$v_1 + v_3 + v_5 = 0$$

$$v_4 + v_5 = 0$$

$$v_1 = v_2$$

- c. Write out the constitutive laws for each component.

$$f_1 = k_1 x_1$$

$$f_3 = b_3 v_3$$

$$f_4 = b_4 v_4$$

$$f_5 = b_5 v_5$$

- d. How many state variables does this system have? What are they?

3 state variable (one mass, two springs)

$$x_1, \dot{x}_1, x_5$$

e. Derive the state equations for this system. Write it out in matrix form.

$$-f_3 - f_4 + f_5 = 0$$

$$-b_3 v_3 - b_4 v_4 + k_5 x_5 = 0$$

$$-b_3 v_3 + b_4 v_5 + k_5 x_5 = 0$$

$$b_4 v_5 = b_3 v_3 - k_5 x_5$$

$$b_4 v_5 = b_3(-v_1 - v_5) - k_5 x_5$$

$$(b_3 + b_4)v_5 = b_3(-v_1) - k_5 x_5$$

$$v_5 = -\frac{b_3}{b_3 + b_4}(v_1) - \frac{k_5}{b_3 + b_4}x_5$$

$$-f_1 + f_3 = m_2 a_2$$

$$-k_1 x_1 + b_3 v_3 = m_2 a_2$$

$$-k_1 x_1 - b_3 v_1 - b_3 v_5 = m_2 a_2$$

$$-k_1 x_1 - b_3 v_1 + b_3 \left(\frac{b_3}{b_3 + b_4}(v_1) + \frac{k_5}{b_3 + b_4}x_5 \right) = m_2 a_2$$

$$-k_1 x_1 - b_3 v_1 + \frac{b_3^2}{b_3 + b_4}(v_1) + \frac{b_3 k_5}{b_3 + b_4}x_5 = m_2 a_2$$

$$a_2 = -\frac{k_1}{m_2}x_1 - \frac{b_3}{m_2}v_1 + \frac{b_3^2}{m_2(b_3 + b_4)}(v_1) + \frac{b_3 k_5}{m_2(b_3 + b_4)}x_5$$

$$\begin{bmatrix} \dot{x}_1 \\ \ddot{x}_1 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{k_1}{m_2} & -\frac{b_3}{m_2} + \frac{b_3^2}{m_2(b_3 + b_4)} & \frac{b_3 k_5}{m_2(b_3 + b_4)} \\ 0 & -\frac{b_3}{b_3 + b_4} & -\frac{k_5}{b_3 + b_4} \end{bmatrix} \begin{bmatrix} x_1 \\ \dot{x}_1 \\ x_5 \end{bmatrix}$$

f. For each statement below, mark T if the statement is true and F if the statement is false. A statement is false if its not always true.

a. If $v_2 > -$, then $f_3 - f_1 > 0$ True

b. If $x_5 > 0$, then $v_4 > 0$ False

c. If $x_1 > 0$, then $v_3 > 0$ False

2. Converting the following ODEs into systems of first-order ODEs by change of variables

a.

$$x = u_1$$

$$\dot{x} = u_2$$

$$\dot{u}_1 = u_2$$

$$\dot{u}_2 = -\sin(t) u_2$$

b.

$$u_1 = x$$

$$u_2 = \dot{x}$$

$$u_3 = \ddot{x}$$

$$\dot{u}_1 = u_2$$

$$\dot{u}_2 = u_3$$

$$\dot{u}_3 = -u_2 u_3$$

c.

$$u_1 = x$$

$$u_2 = \dot{x}$$

$$u_3 = \ddot{x}$$

$$\dot{u}_1 = u_2$$

$$\dot{u}_2 = u_3$$

$$\dot{u}_3 = -t^2 u_3 + u_1 + 3t$$

3. Complex numbers and Euler's formula

- a. Use Taylor series to expand $\sin(\omega t)$ and $\cos(\omega t)$. Write out the first 5 terms in the series expansion, and compare them with the real and imaginary part of $e^{i\omega t}$ in the lecture notes.

Assuming the Taylor series about 0

$$\sin(\omega t) \approx \sin(\omega 0) + \frac{\omega \cos(\omega 0)}{1} t - \frac{\omega^2 \sin(\omega 0)}{2} t^2 - \frac{\omega^3 \cos(\omega 0)}{6} t^3 + \frac{\omega^4 \sin(\omega 0)}{24} t^4$$

$$\cos(\omega t) \approx \cos(\omega 0) - \frac{\omega \sin(\omega 0)}{1} t - \frac{\omega^2 \cos(\omega 0)}{2} t^2 + \frac{\omega^3 \sin(\omega 0)}{6} t^3 + \frac{\omega^4 \cos(\omega 0)}{24} t^4$$

$$\begin{aligned} e^{i\omega t} &= 1 + (i\omega t) + \frac{(i\omega t)^2}{2!} + \frac{(i\omega t)^3}{3!} + \frac{(i\omega t)^4}{4!} + \frac{(i\omega t)^5}{5!} + \dots \\ &= 1 + \underbrace{i\omega t}_{-} - \frac{(\omega t)^2}{2!} - \underbrace{i \frac{(\omega t)^3}{3!}}_{-} + \frac{(\omega t)^4}{4!} + \underbrace{i \frac{(\omega t)^5}{5!}}_{-} + \dots \\ &= \left(1 - \frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} + \dots \right) + i \left(\omega t - \frac{(\omega t)^3}{3!} + \frac{(\omega t)^5}{5!} + \dots \right) \\ &\quad \underbrace{\hspace{10em}}_{\text{Real part}} \quad \underbrace{\hspace{10em}}_{\text{Imaginary part}} \\ &\quad \hspace{10em} \cos(\omega t) \hspace{10em} \sin(\omega t) \end{aligned}$$

The Taylor series expansions are identical to the imaginary and real parts of the Euler identity, respectively.

- b. [5 pts] Consider a complex number $z = z_1/z_2$, where $z_1 = a + ib$ and $z_2 = c + id$. Show that the modulus of z is the quotient of the moduli of z_1 and z_2 ; Show that the argument of z is equal to the difference of the arguments of z_1 and z_2 .

$$\frac{a + ib}{c + id} = \frac{a + ib}{c + id}$$

$$\frac{(a + ib)(c - id)}{c^2 + d^2} = \frac{a + ib}{c + id}$$

$$\frac{ac + bd + i(bc - ad)}{c^2 + d^2} = \frac{a + ib}{c + id}$$

$$\frac{\sqrt{a^2c^2 + 2abcd + b^2d^2 + (a^2d^2 - 2abcd + b^2c^2)}}{c^2 + d^2} = \frac{\sqrt{a^2 + b^2}}{\sqrt{c^2 + d^2}}$$

$$\frac{\sqrt{a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2}}{c^2 + d^2} = \frac{\sqrt{a^2 + b^2}}{\sqrt{c^2 + d^2}}$$

$$\frac{\sqrt{a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2}}{c^2 + d^2} = \frac{\sqrt{a^2 + b^2} \sqrt{c^2 + d^2}}{\sqrt{c^2 + d^2} \sqrt{c^2 + d^2}}$$

$$\frac{\sqrt{a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2}}{c^2 + d^2} = \frac{\sqrt{a^2c^2 + a^2d + b^2c^2 + b^2d^2}}{c^2 + d^2}$$

argument

$$\tan^{-1}\left(\frac{bc - ad}{ac + bd}\right) = \tan^{-1}\left(\frac{b}{a}\right) - \tan^{-1}\left(\frac{d}{c}\right)$$

$$\tan^{-1}\left(\frac{bc - ad}{ac + bd}\right) = \tan^{-1}\left(\frac{\frac{b}{a} - \frac{d}{c}}{1 + \frac{bd}{ac}}\right)$$

$$\tan^{-1}\left(\frac{bc - ad}{ac + bd}\right) = \tan^{-1}\left(\frac{ac\frac{b}{a} - ac\frac{d}{c}}{ac + bd}\right)$$

$$\tan^{-1}\left(\frac{bc - ad}{ac + bd}\right) = \tan^{-1}\left(\frac{bc - ad}{ac + bd}\right)$$

(c) [5 pts] Let $z = Ae^{i\theta}$, deduce that $dz = izd\theta$. Explain the geometric meaning of this expression on the complex plane.

c.

$$z = Ae^{i\theta}$$

$$dz = iAe^{i\theta}d\theta$$

$$dz = izd\theta$$

When z lengthens, both sides of the equation length by the change in z. the RHS is multiplies by i so because changes in $zd\theta$ happen at 90 degrees to z when only theta is changing.

(d) [5 pts] Find the magnitudes and directions of the vectors $2 + i\sqrt{3}$ and $(2 - i\sqrt{3})^2$.

d.

Answers given in: magnitude; direction

$$2 + i\sqrt{3}$$

$$\sqrt{4 + 3} = \sqrt{7}; \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$(2 - i\sqrt{3})^2 = 4 - 3 - i4\sqrt{3}$$

$$\sqrt{1+48} = 7; \tan^{-1}\left(\left(\frac{4\sqrt{3}}{1}\right)\right)$$

(e) [5 pts] Given that both $A + B$ and $i(A - B)$ are real numbers. Show that A and B are complex conjugates. This ensures that the solutions to the oscillation problems are always real-valued.

Hint: Assume $A = x + iy$ and $B = u + iw$.

e.

By definition, if A and B are complex conjugates, then $y = -w$

If

$$A + B = u + x + i(y + w) = \text{real_num}$$

Then

$$i(y + w) = 0$$

$$y + w = 0$$

$$y = -w$$

(e) [5 pts] Use Euler's formula to show

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \text{ and } \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

f.

Display the above relationships geometrically by means of vector diagrams in the complex plane.

$$\frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\frac{\cos(\theta) + i\sin(\theta) + \cos(-\theta) + i\sin(-\theta)}{2}$$

$$\frac{\cos(\theta) + i\sin(\theta) + \cos(\theta) - i\sin(\theta)}{2}$$

$$\frac{2 \cos(\theta)}{2} = \cos(\theta)$$

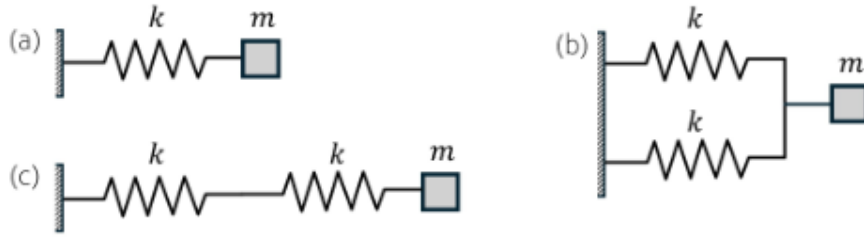
$$\frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\frac{\cos(\theta) + i\sin(\theta) - \cos(-\theta) - i\sin(-\theta)}{2i}$$

$$\frac{\cos(\theta) + i\sin(\theta) - \cos(\theta) + i\sin(\theta)}{2i}$$

$$\frac{isin(\theta) + isin(\theta)}{2i} = \sin(\theta)$$

4. Consider the mass spring system a :



- a. Obtain its equation of motion using force balance, continuity and constitutive laws

$$\begin{aligned} -f_k &= ma_m \\ -kx_k &= ma_m \\ -kx &= ma \text{ (displacements equal.)} \\ \ddot{x} &= -\frac{k}{m}x \end{aligned}$$

- b. Plug in the exponential solution into the equation of motion and obtain the natural freq.

$$\begin{aligned} \lambda^2 x_0 e^{\lambda t} &= -\frac{k}{m} x_0 e^{\lambda t} \\ \lambda^2 &= \omega^2 = -\frac{k}{m} \\ \omega &= \sqrt{-\frac{k}{m}} \end{aligned}$$

- (c) [3 pts] Now, consider the systems in subfig.b and c. Use the results of part (b) to obtain the natural frequencies. Discuss the effect of stiffness on the natural frequency.

c.

$$\begin{aligned} b: \lambda^2 x_0 e^{\lambda t} &= -\frac{k_1}{m} x_0 e^{\lambda t} - \frac{k_2}{m} x_0 e^{\lambda t} \\ \lambda^2 &= \omega^2 = -\frac{k_1 + k_2}{m} \\ \omega &= \sqrt{-\frac{k_1 + k_2}{m}} \\ c: \lambda^2 x_0 e^{\lambda t} &= -\frac{\frac{1}{k_1} + \frac{1}{k_2}}{m} x_0 e^{\lambda t} \end{aligned}$$

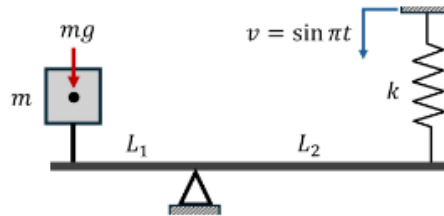
$$\omega = \sqrt{-\frac{\frac{1}{k_1} + \frac{1}{k_2}}{m}}$$

For the series and normal springs, increasing spring constant increases natural frequency, while in the series spring system it decreases natural frequency.

Problem 5 - Forced Oscillation with Transformer

This problem uses MATLAB.

Consider the following system:



5.

- a. Write out Newton's laws without plugging in constitutive laws for this system

$$-F_g L_1 + L_2 F_{kL_2} = L_1 m a_m$$

- b. Write out all the kinematic constraint equations.

$$\frac{x_m}{L_1} = -\frac{x_k}{L_2}$$

- c. Derive the state equation for this system. Write it out in matrix form.

$$-mgL_1 + L_2 k x_k = L_1 m a_m$$

$$a_m = -g + \frac{L_2 k}{L_1 m} x_k$$

$$\ddot{x}_m = -g + \frac{L_2 k}{L_1 m} x_k$$

$$\ddot{x}_m = -g + \frac{L_2 k}{L_1 m} \left(-\frac{L_2}{L_1} x_m \right)$$

$$\ddot{x}_m = -g - \frac{L_2^2 k}{L_1^2 m} x_m$$

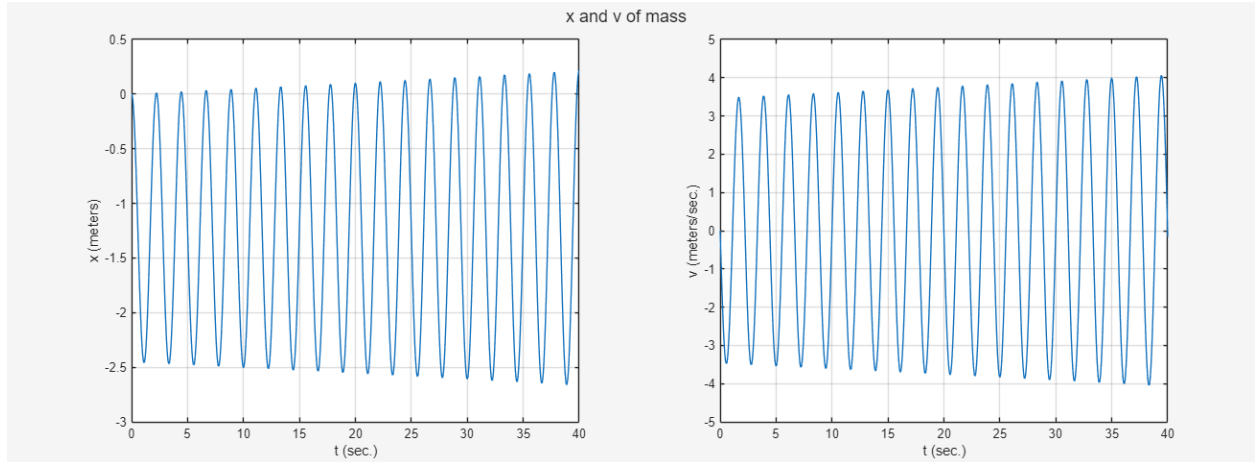
$$\ddot{x} = -g - \frac{L_2^2 k}{L_1^2 m} x$$

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{L_2^2 k}{L_1^2 m} & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ -g \end{bmatrix}$$

- d. Assume the spring is at its natural length when the level is in a horizontal position. Calculate the spring's extension at static equilibrium using the following constants: $m = 0.5\text{kg}$, $g = 9.81$, $k = 1$, $L_1 = 1\text{m}$, $L_2 = 2\text{m}$.

$$\begin{aligned}
 -mgL_1 + L_2 k x_k &= L_1 m a_m \\
 -mgL_1 + L_2 k x_k &= 0 \\
 x_k &= \frac{mgL_1}{L_2 k} = \frac{0.5 * -9.81 * 1}{2 * 1} = -\frac{9.81}{4}
 \end{aligned}$$

- e. Suppose the system starts with $x_s(0) = 0$ and $v_m(0) = 0$. Use the constants in part d and euler integration to approximate the solution for $t = 0 < t < 40$. Plot x and v .



- f. Repeat part e with $v = \sin(2.8t)$ compare results.

